Operation method and application scenarios of AILEIKE ALKB201 fine pressure light pump

The AILEIKE ALKB201 micro-pressure light pump is a portable micro-pressure generating device specifically designed for the verification and pressure calibration of instruments in the micro-pressure range. The device features smooth pressure rise and fall, fine adjustment, good sealing, effortless operation, and portability. It meets the needs of both precision calibration in the laboratory and mobile operations on-site, making it a commonly used calibration device in the field of micro-pressure measurement.

1. Technical indicators

name	Fine pressure light pump	Pressurization method	manual
model	ALKB201	medium	clean air
pressure range	-6000~6000Pa	Overall dimensions	220×120×90mm
Adjust fineness	Min 1 Pa	weight	1.8kg
Output port	2 pieces	Pressure connection	Hose connector thread (M20*1.5)

II. Operation method

First, place the micro-pressure light pump on a flat workbench (in a non-corrosive, vibration-free, and dust-free environment).
Install standard meters and the verified micro pressure (differential pressure) instruments (pressure transmitters, micro pressure gauges, pressure sensors, etc.).

Positive pressure verification process

1. Voltage boosting process

First, turn the ALKB201 handle counterclockwise until it stops. Then, turn the precision adjustment knob counterclockwise to the middle position. Next, close the pressure relief valve. After that, use the handle to gradually increase pressure clockwise until it reaches around the verification value. Finally, use the precision adjustment knob to adjust clockwise to the verification value.

2. Decompression process

First, use the handle to slowly reduce the pressure counterclockwise, stopping when the pressure reaches around the verification value. Then, use the precision fine adjustment to adjust it to the verification value.

3. Pressure relief process

Open the pressure relief valve, gradually release the pressure inside the pipeline, and when the pressure relief reaches 0, the measurement is complete.

Negative pressure verification process

1. Voltage boosting process

First, turn the ALKB201 handle clockwise until it stops, then turn the precision adjustment clockwise to the appropriate position. Next, close the pressure relief valve, and then slowly increase the pressure counterclockwise using the handle until it reaches around the verification value. Finally, use the precision adjustment to adjust counterclockwise to the verification value.

2. Decompression process

First, use the handle to gradually decrease the pressure clockwise, stopping when the pressure reaches around the verification value. Then, use the precision fine adjustment to adjust it to the verification value.

3. Pressure relief process

Open the pressure relief valve, gradually release the pressure in the pipeline, and when the pressure relief reaches 0, the measurement is complete.

III. Application Scenarios

1. Instrument verification in metrology laboratory

Used for periodic verification and calibration of instruments such as micro-pressure capsule pressure gauges, differential pressure gauges, and micro-differential pressure transmitters in metrology and testing laboratories and other places.

2. Industrial site equipment inspection

Applied to industrial scenarios such as chemical, power, and metallurgical industries, for daily inspection, troubleshooting, and on-site calibration of pressure instruments.

3. Scientific research, teaching, and practical training

Used for practical teaching in electromechanical, measurement and control, chemical engineering, and other fields in universities and vocational colleges, as well as for generating micro air pressure and conducting comparative experiments in fluid mechanics laboratories.

IV. Precautions

1. When applying or reducing pressure with the ALKB201 fine pressure light pump, it is necessary to do so slowly to avoid pressure shock.

2. After the operation is completed, the pressure must be released to zero before the inspected instrument can be removed.

3. Regularly check the sealing of sealing rings and joints, promptly replace aging and damaged parts, and ensure the air tightness and calibration accuracy of the equipment.

4. If not in use for an extended period, the equipment should be stored in a dry environment free from corrosive gases;

5. Avoid using it in environments with highly corrosive gases and dust particles.

V. Conclusion

The AILEIKE ALKB201 micro-pressure light pump, with its micro-pressure adjustment capability, lightweight and portable design, and stable operational performance, is suitable for the calibration, verification, and debugging of various micro-pressure instruments. It is easy to operate and widely applicable in various

scenarios, including metrology and testing, industrial production, scientific research, and teaching.